

FINAL PROJECT OVERVIEW

Team Policy:

Undergraduate students may complete the final project either individually or in groups of up to 3 students. In a group, everyone must participate equally. If unequal participation is found, the entire group will be penalized.

Graduate students are required to complete the project individually.

Big idea

You will choose a real dataset (or combine multiple related datasets) and build a well-designed relational database around it. You'll start with a messy, conceptual design and refine it into a normalized schema. Undergraduate projects are expected to include at least 1,000 total rows, while graduate projects must successfully load at least 10,000 total rows. You will also demonstrate SQL mastery through a large set of English-to-SQL queries. Finally, you'll present what you built, what broke, and what you learned.

Dataset rules

- Must be publicly available and cited (link in your submission).
- Must produce a relational model with 6–10 tables after normalization (typical).
- Must have enough complexity to support interesting queries (not just one table).
- You may combine datasets if they share a key or can be linked cleanly (e.g., movies + ratings, transit + stops + schedules, sports + games + players, etc.).
- You must include a short data dictionary: what each table represents and what key fields mean.

Recommended sources (students can pick one):

- Kaggle datasets
- data.gov (US government open data)

- city open data portals (e.g., Portland)
 - public APIs with downloadable dumps
 - IMDb datasets, OpenFlights, open transit feeds (GTFS), etc.
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Undergrad vs Grad difficulty

Undergrad expectations

- 30 English questions + SQL solutions
- At least:
 - 15 basic queries (filters, sorts, joins)
 - 15 advanced (GROUP BY/HAVING, multi-joins, subqueries, 1 view)

Grad expectations (adds depth + performance)

Everything above **plus**:

- 40 English questions + SQL solutions
- Must include:
 - 10 basic queries (filters, sorts, joins)
 - 15 intermediate (GROUP BY/HAVING, multi-joins, subqueries, 1 view)
 - 15 advanced (CTEs, subqueries, more views if needed, materialized views, query tuning like indexing, Stored Procedure/Functions)
- Must include **constraints beyond PK/FK** (at least 3), such as:
 - CHECK constraints (ranges, non-negative, valid statuses)
 - UNIQUE constraints where appropriate

- NOT NULL where appropriate
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The 4 Deliverables

Deliverable 1

1.1. Select a subject area on which you wish to build a database. Write one or two paragraphs that gives a general description / background information on that subject area.

1.2. **For Undergrad:** Go through your dataset and submit your 20 questions in Plain English that you can generate from the dataset.

For Grad: Go through your dataset and submit your 30 questions in Plain English that you can generate from the dataset.

These English questions should reflect realistic questions that someone might ask about the domain. You will translate the questions from this part into queries of the project data. You are permitted to revise these questions later if needed. It is difficult to predict what questions you will be able to answer until later in the project, but this is a starting point.

1.3. Describe what source you intend to use for data, and how you intend to ingest the data into your database. You should choose a subject area where you can easily get several hundred rows of data. This should be written as another paragraph and should include specific details, such as the website or organization where you will find the data and the format of the data. Include a link to your data source so I can browse it too. You do not want to get to week 10 and then “discover” that you cannot find the data or cannot load it. If you cannot find available data, it is likely time for you to change subject areas.

Deliverable 2

2.1. Produce an ER diagram for your domain, and its translation into a relational schema, including all primary keys, unique keys, and foreign keys. You should aim for a database with 6 – 10 tables and at least 1 view. You must submit evidence that you have created at least one table from your schema and populated it with at least one row. You will create your tables in a new schema for the project.

2.2. Load real data into your database. I have a VERY strong preference that you find your data from actual resources. Do not just mock all your data. You must show 5 rows of data for each table and view and row counts. Anticipate a row count of your database (sum of all rows from all tables) to be at least 1,000 for undergrad and 10,000 for grad. If at the end I find you mocked all your data, it will adversely affect your grade.

Deliverable 3

3.1. Express your questions as SQL. At this point you may need to revise your questions (from 1.2) to map to your available data. It is okay to change your questions after you have found and loaded data. You wrote questions before you had a chance to see if the data could answer the questions. If you need to replace a question, state why you are replacing it and provide the new question.

3.2. If all of your questions turn out to be just simple single table queries, you either have questions that are too simple or you did a poor job normalizing your data. This will adversely affect your grade. You should have MANY queries that involve multiple tables in joins. I expect to see joins of 4-6 tables to answer the questions for your project. The required view should join at least 4 tables.

Deliverable 4

4.1. Create a final writeup of your project. For the most part, this deliverable is the accumulation of the deliverables up to this. The final writeup should include:

4.1.1. The Final ER diagram as you actually implemented it.

4.1.2. Table create statements with primary keys, unique keys, and foreign keys for all tables in your project. This should also include all create view statements (you'll need at least one).

4.1.3. Create Index statements for all indexes you created for your project.

4.1.4. A thorough description of how you populated the database.

4.1.5. For each of your questions, the questions in English, their translation to SQL and the full answer to the query(or the first 10 rows). If you needed to change any of your original questions, list the original questions and why you needed to change or replace them.

4.1.6. A listing of 5 rows from each of your tables and row counts of all tables

Final Presentation (Video)

Length: 8 minutes per team (strictly 8 mins)

Must include:

1. Dataset overview + why you chose it
2. Show initial ERD → final ERD (what changed)
3. Demo a few tables + row counts

4. Demo **2 queries** (one intermediate and one advanced)

5. Discuss:

- hardest part
- what you learned
- how confident you feel with SQL now
- what you would do next with more time

Grading breakdown

- Deliverable 1: 10%
- Deliverable 2: 20%
- Deliverable 3: 20%
- Deliverable 4: 35%
- Final presentation: 15%